Select, but Safely: Relevance-Selected Contexts as Experts for In-Context Learning on Drifting Streams

Anonymous Authors[1]

Abstract

A tabular foundation model (TFM) learns a stream through its context, so the choice of which past rows to condition on is the learning algorithm. A natural choice is relevance selection: at every step, rank recent labelled batches by how well they agree with the newest batch and condition on the most relevant ones, not on the most recent. We find that relevance selection performs very unevenly across streams. On seven development streams it matches the accuracy of MICE, a mixture of in-context experts with a worst-case safeguard, but it loses up to half a point to MICE on some streams and, used alone, up to 4.6 points to a plain window. We show that this variance is not a reason to drop selection but a reason to treat it as an expert. Inserted as one more expert into MICE, unchanged, the selected context inherits MICE's any-time regret bound against the plain window, and empirically it costs at most about 0.15 points against MICE on any development stream while keeping most of its gain where concepts recur. On eighteen fresh synthetic streams with controlled recurrence, a pre-registered test is consistent with a recurrence-based explanation: the gain over MICE grows with how often concepts recur (+0.28, +0.15 and +0.01 points for concepts that appear five times, three times and once), is largest for concepts that are hard to learn, and is never below −0.01 points on any stream (rounded). A frozen evaluation on 22 held-out real-stream prefixes and three backbone seeds passes the same no-negative-effect criterion on every seed: MICE-DUO is at most 0.141 points below MICE or FIFO on any stream and is +0.05 to +0.08 points above MICE on average, so its value on real streams is safety at a small gain. Random and oracle controls separate the ranking from the extra expert: ranking retrieves the recurring concept (92% of the selected batches against 31% at random) and, alone, is 34 to 38 points more accurate than a random context, but inside MICE this does not become an established gain over a random extra expert. On the synthetic streams the accuracy advantage (+0.28 to +1.08 points) is not matched in log-loss; on the held-out prefixes a random extra expert lifts MICE as much as the ranked one (+0.13 against +0.07 points).

1. Introduction

Tabular foundation models such as TabPFN (Hollmann et al., 2025) and TabICL (Qu et al., 2025) predict in one forward pass given a context of labelled rows (Müller et al., 2022; Nagler, 2023). On a data stream they replace incremental learners: the model is frozen and the context is updated (Lourenço et al., 2026). What the model knows about the current concept is then exactly the set of rows in its context, and the stream learner reduces to a rule for choosing that set.

Two rules are in use. A window conditions on the most recent rows; it is safe but forgets a concept as soon as it leaves the window. A memory keeps past segments as stored contexts and decides which to recall; MICE (Anonymous, 2026) mixes windows and stored segments under a two-level meta expert and guarantees, with no assumption on the stream, that it is never much worse than the window. Both rules choose contexts made of contiguous rows.

A third rule is suggested by the analysis of non-IID batches (Zhao & Cao, 2024): the error of a learner trained on a set of batches decomposes into a term that grows with the discrepancy between those batches and

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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the test batch, and a term that shrinks with their size. For an in-context learner this is the trade-off between a large context and a relevant one. Relevance selection resolves it per step. It ranks the recent labelled batches by the loss of the TFM conditioned on the newest batch, keeps the most relevant ones together with a short recent anchor, and conditions on this non-contiguous context. We call the pair of this context and the window DUO.

The problem. Relevance selection is attractive and unreliable. On the seven development streams of Section 5.1 DUO is as accurate as MICE on average, gains 1.5 points on a stream whose concepts recur, and loses about half a point on two streams where they do not; the selected context alone is 4.6 points below a plain window on one stream (Table 1). The tested repairs of the rule itself fail: a pre-registered search over the size of the recent anchor traded the losses for the gains and passed on no seed (Section 5.2).

Our answer. We do not choose between selection and memory. We insert the selected context as one more expert of MICE, with MICE unchanged. The consequence is immediate on the theory side and measurable on the empirical side. (i) MICE's slow level is a meta expert over two predictions, the window and the fast mixture; what the fast mixture contains does not enter its bound. The any-time discounted regret of the combined learner against the window is therefore at most $2\ln 2$, as for MICE (Corollary 1). (ii) MICE's fast level reweights its experts every batch from their last loss, so a selected context that was clearly worse on the last labelled batch loses most of its weight; the slow level bounds the cumulative cost otherwise. (iii) Against MICE itself there is no guarantee, and we measure it: on the development streams the combined learner is at most about 0.15 points below MICE on any stream and seed and is above it on average (Table 2).

Contributions. (1) Relevance selection of non-contiguous contexts for TFMs on streams, derived from the discrepancy–size trade-off of non-IID batches, and the observation that its performance is very uneven across streams. (2) A plug-in construction that turns any context-selection rule into an expert of MICE with the window safeguard intact. (3) Pre-registered evidence: on development streams the plug-in reduces the losses of selection to within a pre-specified tolerance; on eighteen fresh streams with controlled recurrence its gain over MICE grows with recurrence and with the difficulty of the concept, and no stream loses more than about 0.01 points (Section 5.3); on 22 held-out real-stream prefixes the frozen criterion holds on all three backbone seeds (Section 5.4). Random and oracle controls show that the ranking retrieves the recurring concept, while the gain of MICE-DUO over MICE is not shown to exceed that of a random extra expert (Section 5.5).

2. Setting and Background

Protocol. A stream arrives in batches $X_1, X_2, \ldots$ of B rows; the labels of batch $t-1$ are revealed before batch t is predicted. A frozen TFM $f_\Theta(\cdot \mid X; \mathcal{C})$ predicts class probabilities for the rows of X given a context $\mathcal{C}$ of at most M labelled rows. A stream learner chooses $\mathcal{C}_t$ from the labelled rows of batches $1, \ldots, t-1$, or combines the predictions of several contexts.

MICE. MICE (Anonymous, 2026) keeps a set $\mathcal{E}$ of in-context experts: FIFO windows of 100, 300 and M rows, and up to 12 stored contexts of past segments. We use MICE with fixed segmentation, which closes a segment every 500 labelled rows; Anonymous (2026) also cut segments at detected changes. The fast level of MICE mixes all experts with weights

$$w_{t,k} \propto \exp\bigl(-\eta\, \ell^{\log}_{t-1,k}\bigr), \qquad k \in \mathcal{E}, \tag{1}$$

where $\ell^{\log}_{t-1,k}$ is the log-loss of expert k on the last labelled batch and $\eta = 10$. Its slow level mixes the fast mixture p_t^{fast} with the M-row window p_t^{win} by exponential weights on discounted half Brier scores $\ell_{t,k} \in [0,1]$,

$$\begin{aligned} L_{t,k} &= \gamma L_{t-1,k} + \ell_{t,k}, \qquad v_{t,k} \propto e^{-\eta_2 \gamma L_{t-1,k}}, \\ p_t &= v_{t,0}\, p_t^{\text{win}} + v_{t,1}\, p_t^{\text{fast}}, \end{aligned} \tag{2}$$

with $\gamma = 0.9$ and $\eta_2 = \frac{1}{2}$. In the implementation both discounted losses start at zero, the outer loss is 0.5 times the half Brier score with unit step, and the weights use $\gamma L_{t-1,k}$ (tempered), which is the same rule; all results in this paper use these settings.

Theorem 1 (Anonymous, 2026, any-time safeguard). For K predictions combined by (2) with $\eta_2 \le \frac{1}{2}$, every sequence of labels and predictions, every T and every k: $\sum_{t \le T} \gamma^{T-t}\bigl(\ell_t(p_t) - \ell_{t,k}\bigr) \le \ln K / \eta_2$.

3. Relevance Selection

Motivation. Zhao & Cao (2024) bound the risk of a learner trained on a weighted set of non-IID batches by a term in the discrepancy between each batch and the target distribution, weighted by the batch's share, plus a term that decreases with the effective sample size. Small batches are individually far from the target; a large set of them is close only if they are relevant. A TFM does not accept sample weights, so we implement the weights as a hard selection.

Rule. At step t the candidates are the labelled batches $t-H, \ldots, t-1$ with $H = 100$. A recent anchor $\mathcal{A}_t$ of the last a batches is always kept. Every older candidate b is scored by the mean log-loss of the TFM conditioned on the anchor,

$$s_t(b) = \tfrac{1}{B} \sum_{(x,y)\in b} -\log f_\Theta(y \mid x; \mathcal{A}_t), \qquad (3)$$

a model-oriented discrepancy between batch b and the current concept in the sense of Zhao & Cao (2023). The selected context $\mathcal{C}_t^{\text{sel}}$ is the anchor together with the $M/B - a$ lowest-scoring older batches; its prediction is $p_t^{\text{sel}} = f_\Theta(\cdot \mid X_t; \mathcal{C}_t^{\text{sel}})$. All scores use labels of batches before t only. Early in a stream, when there are at most $M/B - a$ older candidates, all are kept without scoring.

DUO. The original proposal mixes p_t^{sel} with the M-row window by exponential weights on discounted log-losses ($\eta = 2$, $\gamma = 0.5$) and uses $a = 1$ (an anchor of one batch).

Cost. Scoring takes one TFM call with up to $(H-a)B$ query rows, evaluated in chunks; predicting takes one call with B query rows. On the held-out streams the selection expert adds 30% to MICE's logical TFM calls on average (12% to 43%), but its scoring call is large, and the measured time per batch of MICE-DUO is 1.9 times that of MICE on average (1.2 to 3.9; Section 5.4).

4. MICE-DUO: Selection as an Expert

MICE-DUO is MICE with one more expert: $\mathcal{E}' = \mathcal{E} \cup \{\text{sel}\}$, where expert sel predicts p_t^{sel}. Nothing else changes: the fast weights (1) now range over $\mathcal{E}'$, and the slow level (2) still combines the M-row window with the fast mixture. No constant is re-tuned.

Corollary 1 (The safeguard survives any added expert). For every stream, every selection rule and every T, the discounted half-Brier regret of MICE-DUO against the M-row window, and against its own fast mixture, is at most $\ln 2/\eta_2 = 2 \ln 2$.

Proof. Theorem 1 holds for arbitrary predictions $p_{t,k}$. The slow level of MICE-DUO has $K = 2$ predictions, the window and the fast mixture over $\mathcal{E}'$; which experts the fast mixture contains is not used. □

Remark 1 (What is not guaranteed). Corollary 1 compares MICE-DUO with the window, not with MICE. Adding an expert changes the fast mixture, and the fast mixture of MICE is not one of MICE-DUO's slow-level experts, so Corollary 1 says nothing about the difference to MICE. One way to obtain an analogous guarantee against MICE is to include its full output as a third slow-level expert; we do not evaluate this and instead keep MICE unchanged and measure the difference (Section 5.1).

Remark 2 (Down-weighting after an observed loss gap). An expert whose log-loss on the last labelled batch exceeds the best by Δ has fast weight at most $e^{-\eta\Delta}$ by (1); with $\eta = 10$, a selected context that was 0.3 nats worse on the last batch keeps less than 5% of the fast weight on the next. This says nothing about the next batch's loss under arbitrary drift. When no expert is separated, the slow level decides whether following the fast mixture is worth it, which is the case Corollary 1 covers.

Horizon. Relevance selection only sees the last H batches; MICE's stored contexts have no horizon. The two are complementary: selection recovers concepts that recurred within H batches at the granularity of single batches, stored contexts recover older concepts at the granularity of segments.

5. Experiments

Protocol. As MICE: $B = 100$, $M = 1000$, labels one batch late, TabPFN v2 with four estimators. Every comparison is on the same batches with the same backbone seed. All decision rules were written and committed before the corresponding runs; we report every pre-specified variant, including failures. The 22 real streams not used in development are reserved for frozen confirmation. Backbone seeds change model randomness on fixed data and are not independent data replicates.

5.1. Development Streams: Selection Alone and as an Expert

Seven real streams (Elec2 and six streams used in Anonymous (2026)'s evaluation), first 300 batches, backbone seeds 1 and 2. Table 1 reports accuracy. The selected context alone is the least accurate learner on average, yet on Rialto, whose concepts recur, it is 6.32 points above FIFO and 1.24 above MICE (computed before rounding). DUO matches MICE on average but is 0.50 and 0.47 points below it on Airlines and Weather. MICE-DUO keeps the average and shrinks these losses to −0.08 and −0.01 points.

The pre-registered criterion for MICE-DUO asks for no negative effect: on every stream at most 0.30 points below MICE and below FIFO, a mean not below MICE, and a mean log-loss at most 0.005 above MICE's. Gains are not required. Table 2 reports the primary variant (anchor 100) and three secondary ones. The primary variant passes on both seeds, as do the anchor-

Table 1. Accuracy (%) on the development streams, mean over backbone seeds 1–2. sel: selected context alone; last column: MICE-DUO minus MICE. Bold: best per stream.

Stream	FIFO	sel	DUO	MICE	MICE-DUO	Δ
Elec2	81.52	80.90	81.66	81.82	81.93	+0.11
Airlines	68.21	66.82	67.79	68.29	68.21	-0.08
Phishing	94.66	94.58	94.70	94.71	94.73	+0.02
Poker	91.20	86.63	91.47	91.38	91.54	+0.17
Rialto	75.73	82.05	82.27	80.80	81.65	+0.84
Spam	94.41	91.98	94.66	94.34	94.34	+0.01
Weather	81.21	78.44	80.68	81.15	81.14	-0.01
Mean	83.85	83.06	84.75	84.64	84.79	+0.15

300 selection and the DUO mixture as experts; the anchor-500 selection fails on seed 2 (mean −0.05). The criterion was written after seeing DUO's per-stream results on these streams, so this pass is exploratory; it licenses the tests of Sections 5.3 and 5.4, not a claim.

5.2. The Tested Anchor Variants of Selection Alone Fail

Before MICE-DUO we tried to make DUO safe by itself, by enlarging the recent anchor to 300 and 500 rows so that the selection only fills the rest of the context. The rule required, on every backbone seed, a mean gain over MICE of at least 0.5 points, wins on at least 5 of 7 streams and no stream more than 0.3 points below FIFO. No variant passed on any seed. Larger anchors moved DUO towards FIFO: they reduced the worst loss against FIFO (from about −0.55 to −0.19 with anchor 300) and removed the gain on Rialto (from +1.35 to −0.47 against MICE); these figures average backbone seeds 0–2, whereas Table 1 averages seeds 1–2 (Rialto: +1.47). Even an oracle that picks the better of DUO and MICE per stream after the fact is only 0.23–0.28 points above MICE. Table 7 in Appendix A lists every variant and seed.

5.3. When Does Selection Help? Controlled Recurrence

Eighteen fresh synthetic streams from MICE's grid generator, unused by either method: fifteen blocks of 2000 rows, pure real drift (shared inputs, permuted centroid labels), with $K \in \{3, 5, 15\}$ distinct concepts so that each concept appears 5 times, 3 times, or once; 30 or 100 centroids; three data seeds. Pre-registered hypotheses: (H1) on every stream without recurrence, accuracy at most 0.30 points below MICE and at most 0.30 points below FIFO; (H2) the same two bounds on every stream, with mean log-loss at most 0.005 above MICE; (H3) the gain over MICE increases with recurrence ($K = 3 > K = 5 > K = 15$) and is positive on at least 5 of 6 streams at $K = 3$. Before the results, in our run log rather than in the frozen plan, we noted that with blocks of 20 batches and $H = 100$, a concept with $K = 5$ returns exactly at the edge of the selection horizon, and predicted that the gain at $K = 5$ might be close to that at $K = 15$.

Results. Table 3 reports accuracy by recurrence and difficulty. All three hypotheses hold for the primary variant. The mean gain over MICE is +0.28 points when each concept appears five times, +0.15 when it appears three times and +0.01 when it never recurs; at $K = 3$ it is positive on all six streams. No stream is more than about 0.01 points below MICE (−0.010 at worst), every stream is above FIFO, and the mean log-loss is 0.0015 lower than MICE's. Our prediction for $K = 5$ was wrong: its gain is about half of that at $K = 3$, well above $K = 15$, so the selection horizon did not cancel the benefit of recurrence. The gain concentrates on hard concepts: with 100 centroids it is +0.46, +0.26 and +0.02, with 30 centroids +0.10, +0.04 and +0.01. This matches the learning-curve account of Anonymous (2026): recalling rows pays when a concept takes many rows to relearn. The anchor-300 selection, reported as a secondary variant, passes the same checks. The increment is small next to MICE's own gain over FIFO on these streams (6.3 to 14.5 points). The hit rates of Section 5.5 show that the selected batches do belong to the recurring concept.

5.4. Held-Out Real Streams

Following the mechanism test, we froze a confirmation on 22 real-stream prefixes not used to develop MICE-DUO, although they were previously used to evaluate MICE. We use the first 300 complete batches (or all complete batches for shorter streams), with batch 0 reserved for warm-up, and backbone seeds 0, 1 and 2. The primary method and all constants remain unchanged. The decision requires, separately for each seed, that every stream lose at most 0.30 accuracy points against both MICE and FIFO, that mean gain over MICE be non-negative, and that mean log-loss be at most 0.005 above MICE. We report every stream and seed, including failures. The streams form only four source groups (Airlines, Covertype, Poker and INSECTS); source summaries average streams within source before averaging the four sources. Neither the 22 streams nor the 66 stream–seed pairs are independent replications. The evaluation also records the incremental selection cost.

Results. The frozen criterion holds on every seed (Table 5). Over the 22 streams MICE-DUO is +0.060, +0.078 and +0.048 points above MICE on average

Table 2. Pre-registered no-negative-effect check on the development streams. Differences in accuracy points; ΔLL: mean log-loss minus MICE's.

Expert added to MICE	seed	mean vs MICE	wins	worst vs MICE	worst vs FIFO	ΔLL	pass
+sel (anchor 100)	1	+0.13	6/7	-0.15	-0.04	-0.0053	yes
+sel (anchor 100)	2	+0.17	4/7	-0.07	-0.11	-0.0062	yes
+sel (anchor 300)	1	+0.11	6/7	-0.02	-0.01	-0.0038	yes
+sel (anchor 300)	2	+0.10	7/7	+0.01	-0.05	-0.0040	yes
+sel (anchor 500)	1	+0.01	3/7	-0.13	-0.19	-0.0014	yes
+sel (anchor 500)	2	-0.05	2/7	-0.19	-0.10	-0.0005	no
+DUO mixture	1	+0.06	4/7	-0.05	-0.05	-0.0041	yes
+DUO mixture	2	+0.08	4/7	-0.07	-0.16	-0.0047	yes

Table 3. Controlled recurrence: accuracy (%), mean over three data seeds per cell. Δ: MICE-DUO minus MICE; min: worst stream; last column: streams with $\Delta > 0$.

Concept appears	cent.	FIFO	MICE	MICE-DUO	Δ	min	>0
5 times	30	80.16	94.36	94.46	+0.10	+0.05	3/3
5 times	100	73.92	86.88	87.34	+0.46	+0.44	3/3
3 times	30	79.89	94.06	94.11	+0.04	+0.02	3/3
3 times	100	73.54	86.00	86.26	+0.26	+0.11	3/3
once	30	79.13	92.44	92.45	+0.01	-0.01	2/3
once	100	73.90	80.61	80.62	+0.02	-0.01	2/3

for seeds 0, 1 and 2, no stream is more than 0.134 points below MICE or 0.140 below FIFO, five streams per seed are below MICE, and the mean log-loss is 0.0008 to 0.0012 lower. By source (Table 4) the gain is largest on Poker (+0.15) and positive on Airlines and INSECTS (+0.07, +0.06); on Covertype MICE-DUO is consistently but negligibly below MICE (between −0.02 and −0.01 on every seed). Averaged over sources, MICE-DUO is +0.07 points above MICE and +0.50 to +0.59 above FIFO. These results confirm the development finding on unseen streams: selection as an expert does no measurable harm and adds a small gain; a random extra expert adds as much (Section 5.5).

Cost. The selection expert adds 30% to MICE's logical TFM calls on average (12% to 43% across streams), with 571 fits and 2 714 forward calls per stream on average. Its scoring call predicts up to $(H - a)B$ rows, so time grows faster than calls: the measured time per batch of MICE-DUO is 1.9 times that of MICE on average (1.2 to 3.9). These times were measured with up to eight concurrent processes on one GPU and include contention; they describe this run, not an isolated latency benchmark.

Execution. The run used one, then up to eight, then two or three concurrent workers because of intermittent CUDA failures on our machine; each change was recorded as a dated amendment before any score was inspected, the snapshot, tasks, data and evaluator were unchanged, and failed batches were recomputed from checkpoints in new processes. Predictions are bit-identical across concurrency levels.

Table 4. Held-out real-stream prefixes: accuracy (%) per source, mean over three backbone seeds. Δ: MICE-DUO minus MICE (mean, range over seeds); worst: lowest stream–seed difference.

Source	str.	FIFO	MICE	MICE-DUO	Δ	range	worst
Airlines	4	67.73	67.76	67.82	+0.07	[+0.05, +0.09]	-0.04
Covertype	5	93.63	93.93	93.91	-0.02	[-0.02, -0.01]	-0.13
INSECTS	9	79.08	79.99	80.05	+0.06	[+0.05, +0.08]	-0.01
Poker	4	92.17	92.86	93.01	+0.15	[+0.09, +0.21]	-0.12
Sources	22	83.15	83.63	83.70	+0.07	[+0.05, +0.08]	-0.13

Table 5. Frozen held-out criterion per backbone seed (22 streams). Differences in accuracy points; ΔLL: mean log-loss minus MICE's.

seed	mean vs MICE	worst vs MICE	worst vs FIFO	<0	ΔLL	pass
0	+0.060	-0.134	-0.087	5/22	-0.0008	yes
1	+0.078	-0.040	-0.140	5/22	-0.0012	yes
2	+0.048	-0.077	-0.130	5/22	-0.0010	yes

5.5. Is the Gain from Ranking? Random and Oracle Contexts

Two controls replace the ranking of Equation (3) and keep everything else: the same anchor, the same horizon H, the same M/B batches passed in chronological order. Random draws the non-anchor batches uniformly from the older candidates (three draws per stream and backbone seed). Oracle, on the synthetic streams only, takes the most recent candidates of the query batch's concept and fills the remaining slots at random. Each control is evaluated alone and as the extra expert of MICE, combined by the same rule as sel (Section 4). The hit rate is the share of selected non-anchor batches whose concept is that of the query batch. On the synthetic streams the intervals are paired t intervals over the six streams of a level; on the held-out prefixes they are moving-block bootstrap intervals over batches (blocks of 17) conditional on the 22 prefixes, and the four sources do not support a population interval.

Ranking retrieves the recurring concept. On the synthetic streams 92%, 88% and 71% of the ranked batches belong to the query batch's concept for $K = 3$, 5 and 15, against 31%, 21% and 14% for random batches and 96%, 92% and 71% for the oracle (Table 6). Alone, the ranked context is 34 to 38 points more accurate than a random one, with lower log-loss on every stream, and 0.3 to 2.8 points less accurate than the oracle. This is the selection of low-discrepancy batches that motivates the rule (Section 3).

Inside MICE, synthetic streams. With ranking, MICE is more accurate than with a random extra expert on all 18 streams, by +0.32, +0.28 and +1.08 points at $K = 3$, 5 and 15; the interval excludes zero at $K = 3$ and 5 and reaches it at $K = 15$. The log-loss difference is within ± 0.006 at $K = 3$ and 5 with intervals that contain zero, and +0.013 at $K = 15$. By the pre-specified reading, which requires both metrics, a deployed advantage at $K = 3$ is not established. The fast level gives a random context little weight (0.02 to 0.05, against 0.19 to 0.31 for the ranked one), yet without recurrence it still costs MICE 1.07 points, and so does the oracle, which then has few same-concept candidates and is filled at random; the ranked context costs nothing there (+0.01).

Held-out real prefixes. No advantage of ranking is established. Averaged over sources, a random extra expert lifts MICE by +0.13 points and the ranked one by +0.07; the difference is -0.05 $[-0.09, -0.01]$ points in accuracy and -0.008 $[-0.012, -0.006]$ in log-loss, positive on one source (INSECTS) and most negative on Poker (-0.24). Alone, the ranked context is less accurate than a random one on Airlines, Covertype and Poker (-11.8 points on Covertype) and more accurate on INSECTS (+5.4). The small real-stream gain of MICE-DUO over MICE (Section 5.4) is therefore the gain of an extra context expert; on these prefixes it is not shown to come from ranking.

6. Related Work

In-context stream learning. TFMs with sliding memories outperform adaptive tree ensembles on drifting streams (Lourenço et al., 2026), and drifting priors improve them under temporal shift (Helli et al., 2024). MICE (Anonymous, 2026) adds a memory of stored contexts with a worst-case safeguard. Relevance selection is a third way to build a context, at the granularity of batches and within a horizon.

Recurring drift. Pools of models with reuse decisions (Gonçalves & de Barros, 2013; Gama & Kosina, 2014) and the mixture of online and offline experts (Zhao et al., 2025) recall trained models; drift detectors reset them (Gama et al., 2004). See Gama et al. (2014); Lu et al. (2019) for surveys.

Discrepancy and batch weighting. R-divergence measures model-oriented discrepancy (Zhao & Cao, 2023); weighting non-IID batches by discrepancy improves out-of-distribution detection (Zhao & Cao, 2024). Our score (3) is a one-batch model-oriented discrepancy used for selection rather than weighting.

Expert advice. Exponential weights with exp-concave losses (Cesa-Bianchi & Lugosi, 2006; Hazan et al., 2007) make adding an expert cheap in regret; Corollary 1 is an instance at the level of contexts.

7. Limitations

The gain of MICE-DUO over MICE is small: +0.15 points on the development streams, mostly from one stream, and at most about half a point on the synthetic streams, where MICE itself gains 6.3 to 14.5 points over the window. The synthetic streams are pure real drift with a fixed input distribution, cleaner than real streams. The mechanism test uses one backbone seed (with three independent data seeds), and the development backbone seeds share data. All experiments use MICE with fixed segmentation; the selection expert was not run with segments cut at detected changes. Changing the number of concepts changes both recurrence frequency and return interval, so this experiment does not isolate their individual effects. Real-stream confirmation concerns 300-batch prefixes from four sources. The safeguard is against the window, not against MICE; the small observed losses against MICE satisfy an empirical tolerance criterion, not a guarantee of non-decreasing accuracy. Selection costs extra TFM calls per batch (30% more calls and about twice the time per batch in our held-out run). On held-out real streams the gain over MICE is +0.05 to +0.08 points on average and slightly negative on one source (Covertype), and a random extra expert gains as much; the controls use one backbone seed on the synthetic streams and a single anchor ($a = 1$).

8. Conclusion

Relevance selection gives an in-context learner the right past rows when concepts recur and the wrong ones when they do not. As an expert of a mixture with a window safeguard, the first property is kept and the second is contained: on these controlled streams the gain is larger for frequent recurrence and harder concepts, and is approximately zero on average without recurrence. Individual losses remain within the pre-

Table 6. Ranked, random and oracle contexts. Hit: share (%) of selected non-anchor batches from the query batch's concept. Alone: accuracy of the ranked minus the random context (points). vs MICE: accuracy of MICE with the extra expert minus MICE. Deployed: MICE with the ranked minus MICE with the random expert, accuracy (points) and log-loss, mean and 95% interval (synthetic: paired t over six streams; held-out: conditional block bootstrap, source averages for the other columns). Positive values favour ranking.

	Hit			Alone	vs MICE			Deployed: ranked − random	
Concept appears	rank	rand	orc	Δacc	rank	rand	orc	Δacc	Δlog-loss
five times ($K = 3$)	92	31	96	+34.0 [+32.5, +35.4]	+0.28	−0.04	+0.07	+0.32 [+0.16, +0.49]	−0.006 [-0.018, +0.007]
three times ($K = 5$)	88	21	92	+37.7 [+36.7, +38.7]	+0.15	−0.13	−0.03	+0.28 [+0.20, +0.36]	+0.001 [-0.006, +0.008]
once ($K = 15$)	71	14	71	+35.7 [+34.2, +37.2]	+0.01	−1.07	−1.03	+1.08 [-0.00, +2.17]	+0.013 [+0.002, +0.025]
held-out, 4 sources	–	–	–	−2.6	+0.07	+0.13	–	−0.05 [-0.09, -0.01]	−0.008 [-0.012, -0.006]

specified tolerance. On 22 held-out real-stream prefixes this holds on every backbone seed: no measurable harm and a small gain. The ranking itself finds the recurring concept, close to an oracle, but MICE's mixture does not turn this into an established gain over a random extra expert; on real streams the gain is that of one more context expert.

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A. Standalone Selection with Larger Anchors

Table 7. Pre-registered search over the recent anchor for standalone DUO on the seven development streams (Section 5.2). Criterion: mean gain over MICE at least +0.50 points, at least 5 of 7 wins, and no stream more than 0.30 points below FIFO, on every seed. Anchor in rows; safe mixture: half-Brier, $\gamma = 0.9$, tempered weights. Seed 0 was exploratory; the safe mixture with anchor 100 was not run on seed 0. The criterion was fixed before the runs in the overnight run plan (experiments/night_20261007/README.md, item 5).

seed	anchor	method	mean vs MICE	wins	worst vs FIFO	pass
0	100	DUO	+0.02	3/7	-0.55	no
0	300	DUO	-0.12	2/7	-0.19	no
0	300	safe mixture	-0.22	3/7	-0.21	no
0	500	DUO	-0.60	2/7	-0.44	no
0	500	safe mixture	-0.58	2/7	-0.46	no
1	100	DUO	+0.10	4/7	-0.47	no
1	100	safe mixture	-0.04	2/7	-0.56	no
1	300	DUO	+0.07	4/7	-0.18	no
1	300	safe mixture	-0.04	4/7	-0.16	no
1	500	DUO	-0.37	3/7	-0.28	no
1	500	safe mixture	-0.44	3/7	-0.74	no
2	100	DUO	+0.11	3/7	-0.58	no
2	100	safe mixture	-0.06	1/7	-0.69	no
2	300	DUO	-0.06	2/7	-0.16	no
2	300	safe mixture	-0.15	2/7	-0.22	no
2	500	DUO	-0.60	2/7	-0.56	no
2	500	safe mixture	-0.63	3/7	-0.80	no